

Assessing the Validity of Airfoil Lift and Drag Simulation Software

Laboratory Report

April 3, 2026

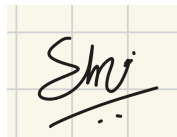
Patrick Jacobs

A handwritten signature in black ink, appearing to read 'Patrick Jacobs', with a stylized flourish at the end.

Steven Ramsey

A handwritten signature in blue ink, clearly legible as 'Steven Ramsey'.

Shrish Rao

A handwritten signature in black ink, appearing to read 'Shri', with a horizontal line and a small mark below it.

Introduction + Hypothesis

Computerized technology has become increasingly central to the facilitation of engineering design, and among the leading reasons is due to the development of digital analysis software. Various programs have been created with the purpose of simplifying design decisions, running simulated analyses, and allowing for rapid efficiency improvements across many disciplines. Aeronautics is no exception, and the development of computational fluid dynamics (CFD) packages and similar computational methods for 2D airfoil properties have been integral to the advancement of the aviation industry. However, as with all simulations, their value lies only in their abilities to predict real-life scenarios with accuracy and precision.

The primary objective of this experiment is to model and draw comparisons between three unique airfoil analysis tools—NASA FoilSim, XFOIL, and ANSYS—to assess their reliability for predicting the lift and drag of *airfoils*. Airfoils are necessary aerodynamic components that directly affect lift, drag, and flow behavior. To do this, each software's lift and drag predictions will be compared with those experimentally produced using wind tunnel tests, evaluating two different airfoils with significant variations in profile (Eppler E423 High-Lift and NACA 2415; Figure 1) to evaluate their aerodynamic abilities. While several research studies have been conducted contrasting XFOIL with ANSYS Discovery, these applications were primarily done using low Reynolds numbers regimes (Gunel et al. 2016, Morgado et al. 2016), whereas this experiment focuses on more extreme turbulent applications. Furthermore, NASA FoilSim has limited assessment in scientific contexts given its nature as an educational tool and not an official solver. However, this is the most easily accessible online tool for airfoil evaluation, so its ability to approximate lift and drag will also be analyzed.



Figure 1. Side profiles of NACA 2415 (top) and Eppler E423 (bottom) airfoil fabricated prototypes.

It is hypothesized that ANSYS Discovery will be the most accurate software for assessing lift and drag (i.e., generated lift and drag will most closely align with wind tunnel test forces), followed subsequently by Xfoil and NASA FoilSim. This prediction is based on both the accessibility and complexity of the models used. ANSYS Discovery is a high-end simulation software, with licenses costing thousands of dollars; its price and reputation make it the leading candidate for conducting reliable, realistic simulations. Xfoil is a 2D airfoil analysis tool with a strong reputation in its respective industry, but it is an open-source software and its two-dimensional nature does not fully capture its real-life application as a 3D shape. NASA FoilSim is simply an education software and not originally intended for use in legitimate design contexts, so it is expected that it only generates “ballpark” estimates.

Procedure + Methods

Procedures for evaluating the lift and drag forces acting on the airfoil were broken down into four parts: experimental results, NASA FoilSim, Xfoil, and ANSYS Discovery.

Experimental Results

The experimental results, which serve as a baseline for comparison of all the computational methods of analysis, were found via tests in the UVA Low-Speed Wind Tunnel apparatus. The tunnel operates by using a fan to draw air into a contracting funnel that accelerates air through the testing section and out a diffuser (Forman 2026). To validate the calibration of the force sensor, a dynamometer, mounted in a way as such to apply a known force, was applied to the force sensor (Figure 2). The readings from the dynamometer and force sensor were similar to within an acceptable range (~ 0.05 lb), so testing could proceed.



Figure 2. Photograph of dynamometer configuration used in calibration of wind tunnel force sensor.

The NACA 2415 and Eppler E423 airfoils were fabricated with ABS plastic via 3D printing courtesy of the Rapid Prototyping Lab (RPL), each with a span of 8" and chord length of 6" to maintain similarity. The testing protocol for each airfoil involved attaching its adapter onto the hex adapter, adjusting the angle to an initial angle of attack of zero (Figure 3), and taring all readings using the data acquisition system's graphical user interface (GUI) (Figure 4). The wind tunnel was then turned on to an initial fan speed of 1000 RPM, and after the tunnel generated a steady airflow, a sample was taken by averaging the lift and drag force readings inside the tunnel over the course of two seconds per sample. One sample was taken at each angle from 0° to 20° , increasing in increments of 5° . The tunnel speed was

then increased to 1250 RPM, and five more measurements were taken. The final five measurements were taken at varying angles with a fan speed of 1500 RPM. Photographs demonstrating the airfoils' testing during the wind tunnel tests are shown in Figure 5.

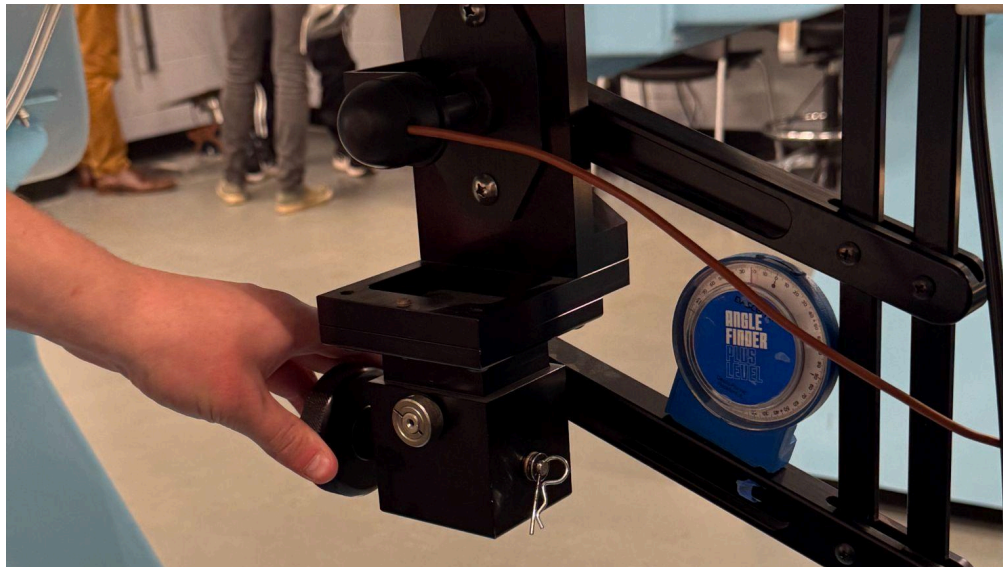


Figure 3. Mechanism for adjusting angle of attack (α) of airfoil in wind tunnel.

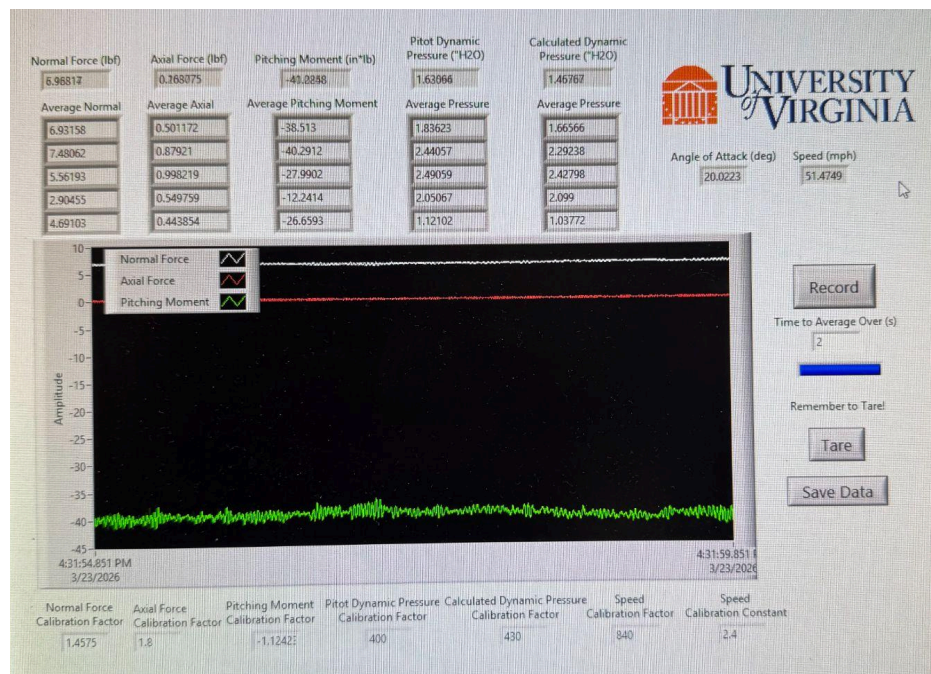


Figure 4. GUI of wind tunnel, displaying essential parameters and facilitating data acquisition.

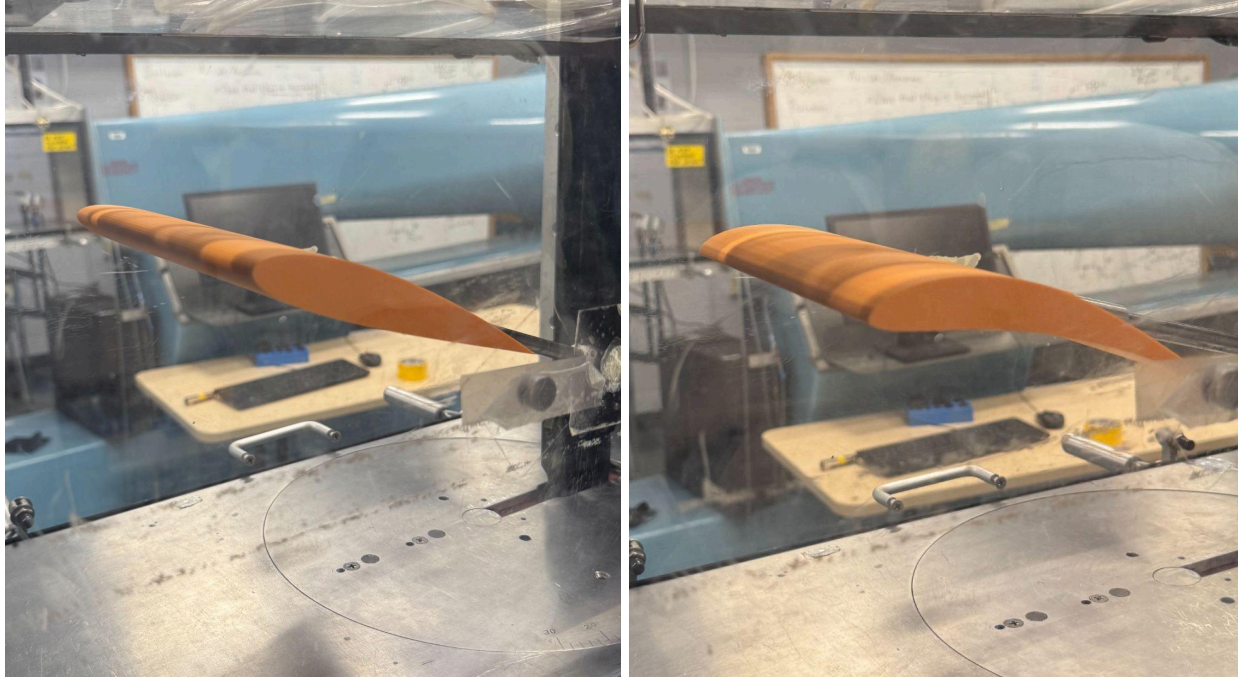


Figure 5. NACA 2415 (left) and Eppler E423 (right) airfoils experiencing testing inside the wind tunnel.

After all trials were conducted, the data was exported into a spreadsheet for analysis. The wind speed (u) in the tunnel was found using the pitot dynamic pressure (q), a quantity measured within the tunnel apparatus. The relationship is defined as $u = \sqrt{\frac{2q}{\rho}}$, where ρ is the density of the fluid. The density of air can be found using the Ideal Gas Law, where $\rho = \frac{PM}{RT}$. The air density for the experiment was found to be 1.197 kg/m^3 based on the measured parameters and constants below:

Table 1. Ideal Gas Law Parameters

Symbol	Parameter	Value
P	Pressure	100.701 kPa
M	Molar Mass of Dry Air	0.02896 kg/mol
R	Universal Gas Constant	8.314 J/(mol*K)
T	Temperature	293.15 K (68°F)

NASA FoilSim

While NASA has several iterations of its interactive airfoil application available online, The NASA FoilSim III Elementary JS Version 1 model was used for this specific analysis. The dashboard of this site has been intentionally designed to be an accessible interface, so all the modifying variables and outputs are visible on one condensed screen. Its six modifying variables are speed, altitude, angle, camber, thickness, and wing area. Speed was found by finding the average velocity the wind tunnel produced across each of the 1000, 1250, and 1500 RPM trials from the pitot probe. Altitude was not a factor, and was assumed to represent the elevation of the flight simulator in order to match the proper atmospheric conditions. Camber and thickness percentages were drawn directly from the airfoil specification sheets, available in the Appendix. Wing area, in reality, was 0.333 square feet (6" x 8"), but since this variable only affects the total lift (and not the coefficients C_L and C_D) and the simulator only outputs lift and drag to the nearest integer, it was modified by a factor of x1000 to obtain a more precise result. For example, in Figure 6, the estimated lift would be "3.030 lbs" to account for the adjustment of the decimal. As with the experimental data, fifteen total measurements were taken—five for each of the three wind speeds, iterating through angles of 0 to 20 degrees. The biggest advantages of FoilSim lie in its ease of access, simplified structure and quick result response with little to no barrier for entry. However, as its name suggests, it is only an "elementary" model, and the sophistication of the model calculations may reflect that.

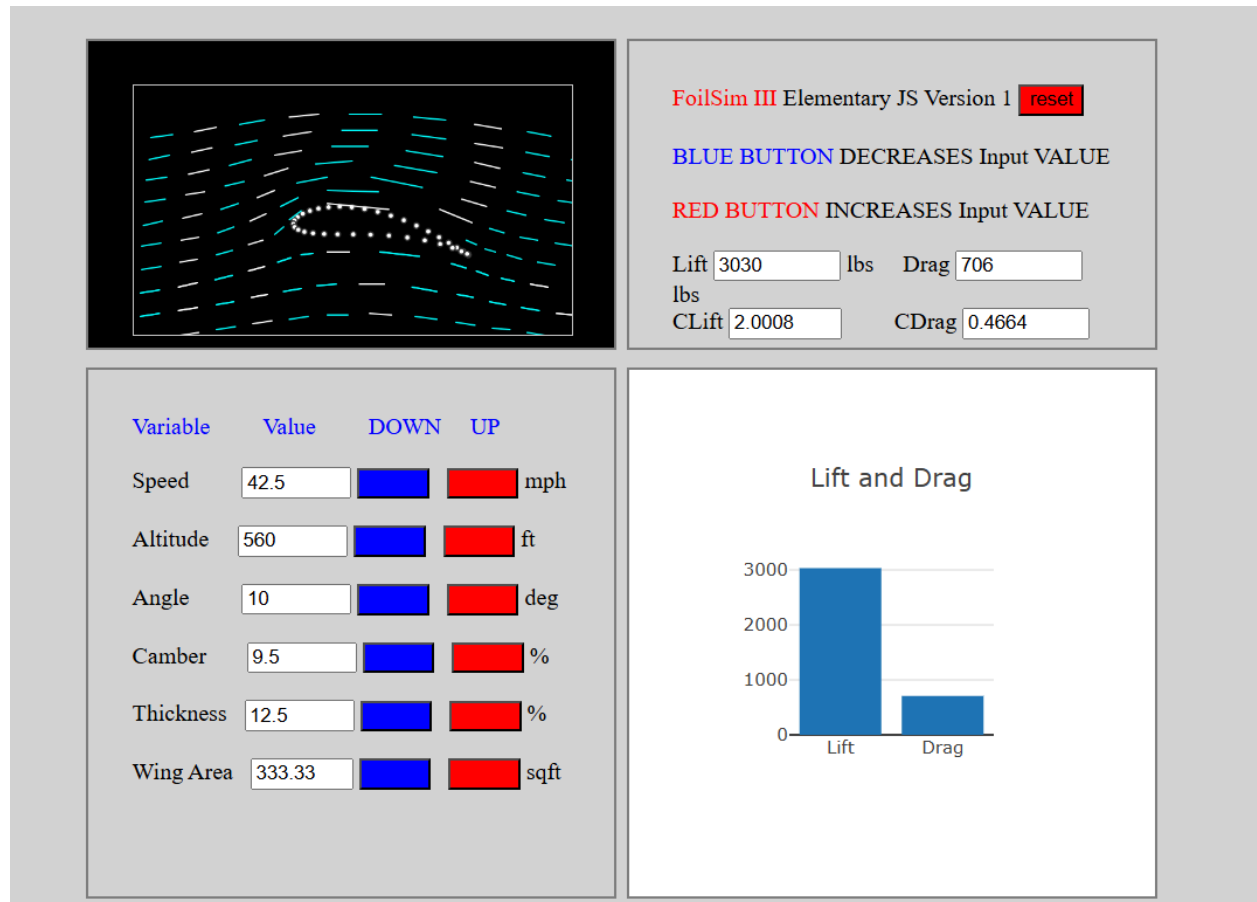


Figure 6. User interface for NASA FoilSim airfoil simulator, displaying modifying variables and outputs.

XFoil

The XFoil program is an open-source 2D airfoil analysis software developed by engineers at the Massachusetts Institute of Technology. Considered to be an industry standard tool, it has high relevance in modern aerodynamic applications and is regularly relied on by engineers as a basis for their decisions. The procedure for conducting an XFoil analysis requires first loading the relevant airfoil's generic data file (.dat) and entering the program's operating window. Because the project aims to model real-life airflow, viscous mode was toggled on. At this point, users are prompted to input the Reynolds number (Re) and Mach number (Ma), two dimensionless values that describe the conditions of the fluid flow.

Reynolds number represents the ratio of inertial and viscous forces acting upon a surface and is defined in airfoil applications as $Re = \frac{\rho v c}{\mu}$ where ρ is the fluid density, v is the freestream velocity, c is the chord length, and μ is the dynamic viscosity (for room temperature air, this can be assumed to be approximately $1.8 \cdot 10^{-5} \frac{kg}{m \cdot s}$). Each of these variables is already known from the “Experimental Results” section and the Reynolds number could therefore be calculated for these applications. Mach number is a non-dimensionalized unit of speed which is normalized to the speed of sound, and is useful to the program when considering compressibility effects. It is defined as $Ma = \frac{v}{a}$, where v is again freestream velocity and a represents the speed of sound (343 meters per second in air at room temperature).

After entering Re and Ma , the user then must automatically input an angle (in degrees), at which point the software will have all variables defined in order to iterate and solve for the coefficients of lift and drag. In events where the window has not converged through ten iterations, it was manually prompted to continue iterating and eventually all values stabilized, output in a visual window shown in Figure 7. Upon obtaining C_L and C_D , the lift and drag forces had to be manually computed since XFOil only considers two dimensions. Defined as $L = \frac{1}{2} C_L \rho v^2 A_p$ and $D = \frac{1}{2} C_D \rho v^2 A_p$, with A_p representing the wing’s planform area, the predicted forces were identified.

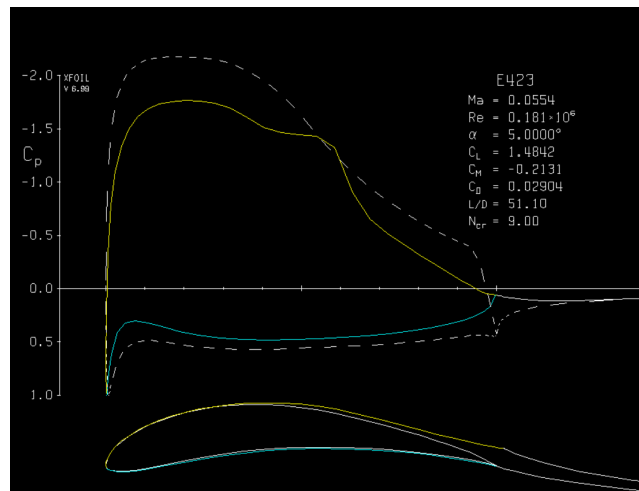


Figure 7. XFOil plot window, including lift and drag coefficients.

ANSYS Discovery

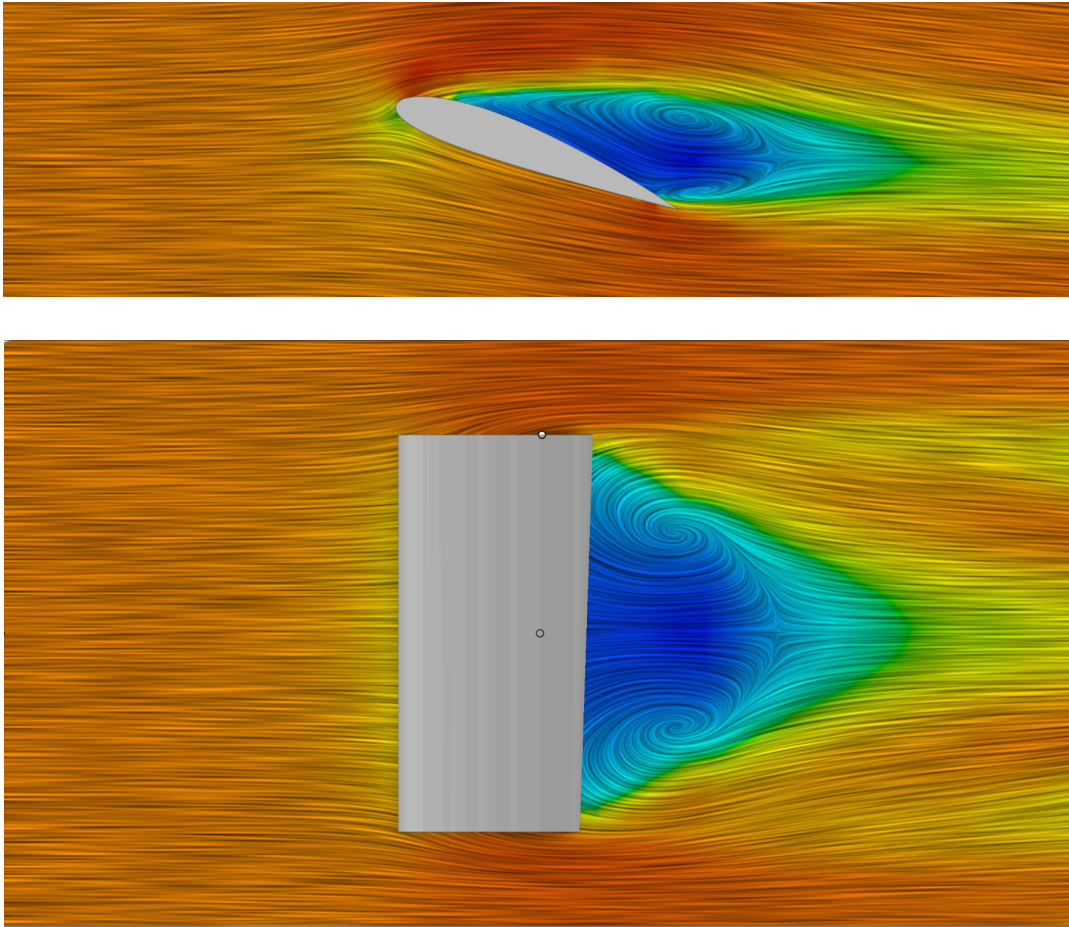


Figure 8. Direction Fields of NACA2415's 20 Degree, 30.4 m/s simulation.

The procedure we used for Ansys Discovery was first importing our geometries for the NACA2415 and Eppler 243 airfoils into their respective files, and creating a fluid flow study within the software, selecting the inlets, outlets, specifying gauge pressure for the outlet and fluid velocity for the inlet. This creates a fluid gradient across an enclosure, creates a ground plane, and creates enclosing walls, however these need to be altered in dimensions to accurately reflect the wind tunnel that was used. The wind tunnel's dimensions are roughly a 0.300 meter square, extruded 0.558 meters long, thus a new enclosure is created of those dimensions centered about the center of our airfoil's chord length and height. Following this, the boundary walls and ground plane must be redefined and the stock enclosure removed.

The airfoil's body is then set to be a “cutter body”, essentially creating a boolean between it and the enclosure to reflect the surfaces of the airfoil. These inverted faces of the enclosure are then selected, and a force monitor in both the Y and Z axes is applied, allowing the total of all forces across the airfoil to be examined when a simulation is run. The simulation can then be run, and the velocity can be altered to get new forces for different wind speeds.

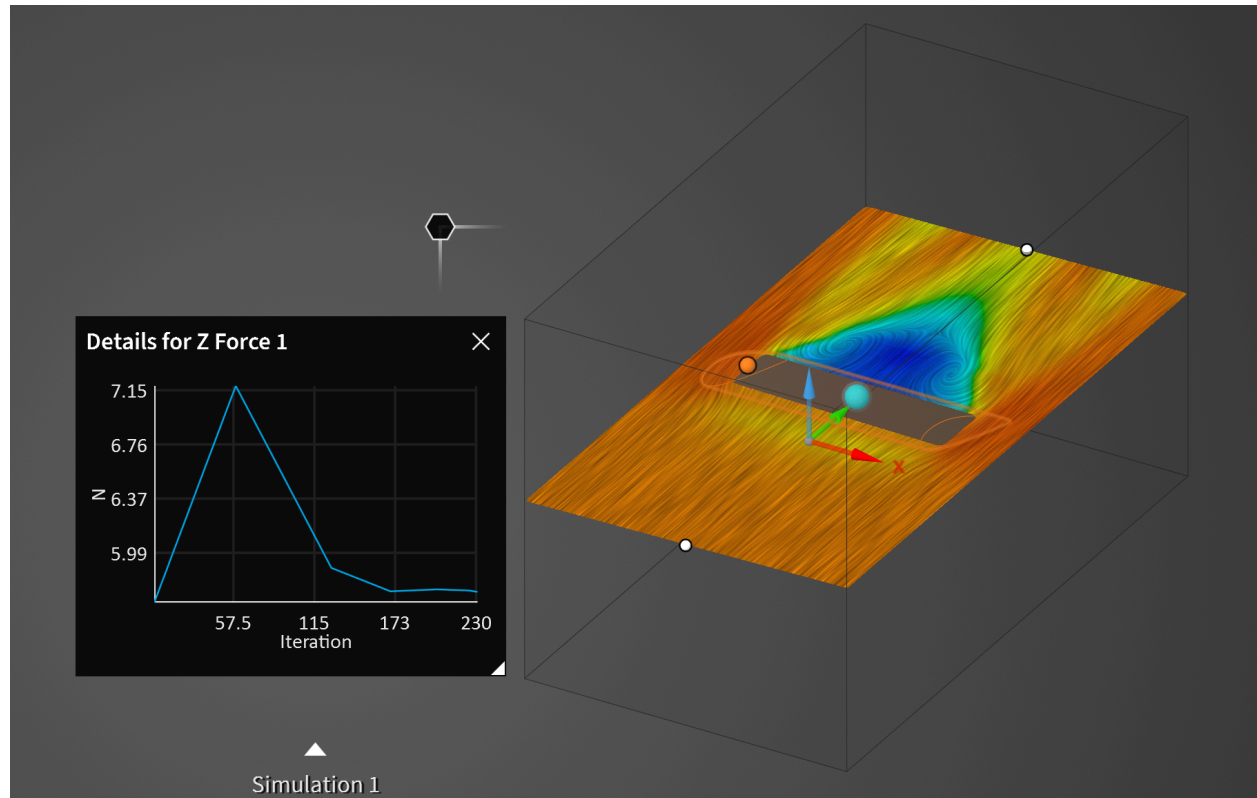


Figure 9. Demonstration of Z Force Monitor (Lift) on the airfoil's inverted faces cut from the Enclosure.

For each new angle of attack, the main body of the airfoil is rotated 5 degrees, and the enclosure's body is recomputed to reflect the change. The same method is used to capture the forces following each shift.

For consistency, simulations were run using just the extruded profile of the airfoil without the attachment apparatus that exists in real life on the model for use in the wind tunnel. This decision was made to align the models used with our other solvers, as NASA FoilSim and XFOIL are only capable of processing the 2D splines of the airfoil. This will undoubtedly introduce error between the real world

model and the Ansys model, but it should result in an equal error across all airfoil models used, allowing us to determine the comparative error between models.

One last thing to discuss is why exactly ANSYS may show better results than the 2D simulators. As you can see in the above figure, the flow of the airfoil changes as you shift across the length of the airfoil, creating a triangular section that shows vortexes behind the airfoil. This additional dimension in which the fluid flow is simulated is expected to give a better result for the forces.

Results + Analysis

Table 2. Lift Pivot

Angle	FoilSim NACA Error (%)	FoilSim Eppler Error (%)	XFoil NACA Error (%)	XFoil Eppler Error (%)	ANSYS NACA Error (%)	ANSYS Eppler Error (%)
0	36.21	68.16	24.74	67.71	70.07	64.58
5	47.58	27.28	50.30	25.28	88.49	54.30
10	26.59	14.08	12.18	7.34	86.66	50.16
15	6.62	17.25	21.86	3.66	80.80	40.42
20	32.18	17.66	37.14	4.58	82.26	27.99

Table 3. Drag Pivot

Angle	FoilSim NACA Error (%)	FoilSim Eppler Error (%)	XFoil NACA Error (%)	XFoil Eppler Error (%)	ANSYS NACA Error (%)	ANSYS Eppler Error (%)
0	50.21	49.60	83.56	80.23	78.38	69.19
5	47.92	26.22	93.04	90.06	97.41	53.91
10	54.95	73.99	86.58	85.46	75.86	31.54
15	593.54	195.32	157.79	65.73	156.46	38.72
20	280.61	268.18	168.13	65.80	151.51	259.99

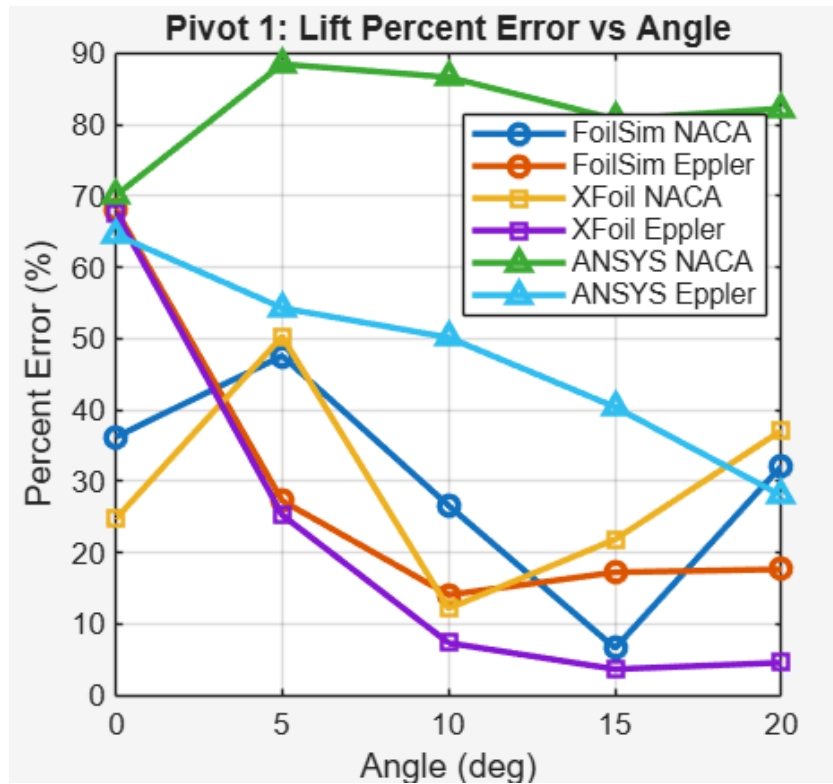


Figure 10. Lift Percent Error vs Angle Graph.

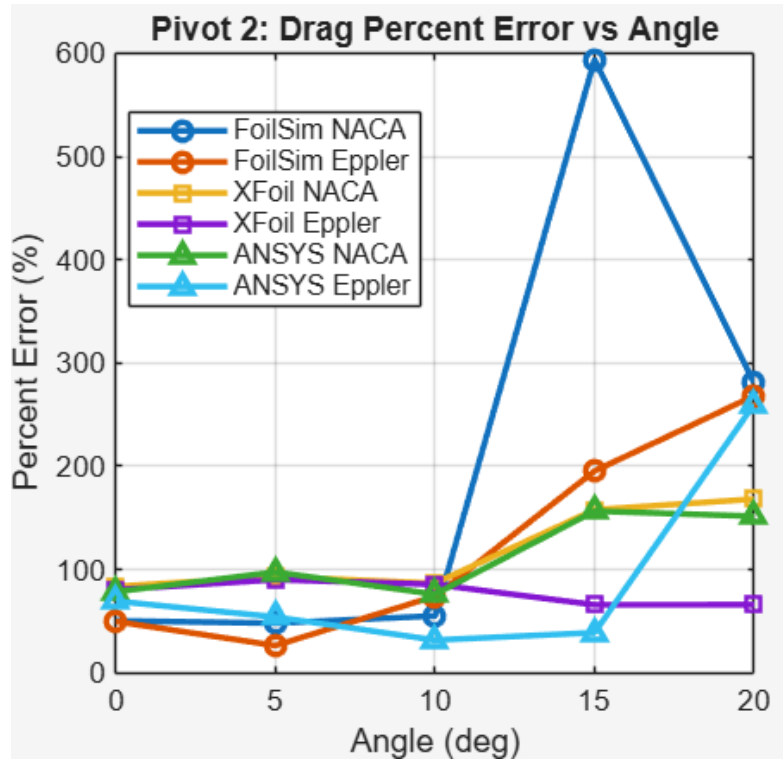


Figure 11. Drag Percent Error vs. Angle Graph.

The results show differences between how each simulation tool predicts lift and drag compared to the experimental wind tunnel data. Overall, lift was predicted more accurately than drag, and errors increased significantly at higher angles of attack.

For the lift data, the models performed well at lower angles (between 0-10) with the percent errors remaining moderate. This lines up because flow over the airfoil is more stable in this region which makes it much easier for models to approximate pressure distributions. At 10°, Xfoil showed low errors of about 12.18% (NACA) and 7.34% (Eppler), while FoilSim had errors of 26.59% and 14.08%, respectively. ANSYS had higher lift errors even at low angles with values like 86.66% at 10°. This can be attributed to setup limitations or modeling simplifications rather than physical inaccuracies. As the angle of attack increased, certain models improved before degrading. At 15°, FoilSim's NACA error dropped to 6.62%, and Xfoil's Eppler error decreased to 3.66% which is in alignment with the experimental data in

this range. At 20° , the XFoil NACA error rose to 37.14% and FoilSim to 32.18% which shows the possibility of flow separation.

For the drag data, the models showed much larger errors across all angles, especially at higher angles of attack. At 0° errors were already high, XFoil showed 83.56% and 80.23%, and ANSYS showed 78.38% and 69.19%. As the angle of attack increased, the error continued to rise significantly. At 15° , FoilSim's NACA drag error increased to 593.54%, and XFoil and ANSYS showed large errors of 157.79% and 156.46%, respectively. These large anomalies show that none of the simulation tools were truly able to capture the highly separated and turbulent flow conditions present at higher angles of attack.

An important observation is the difference between the NACA and Eppler airfoils. For lift, the errors were generally lower in the NACA airfoil compared to the Eppler airfoil at certain angles. At 15° , XFoil percent error was at 21.86% for the NACA airfoil while the Eppler airfoil had an error of only 3.66%. When measuring drag at 20° , ANSYS showed 151.51% error in the NACA airfoil and 259.99% in the Eppler airfoil which again highlights the difficulty in modeling drag forces.

Overall, the results support the initial hypothesis that more advanced simulation tools provide better predictions, but also highlight that all models have limitations. While ANSYS had better physical representation through 3D modeling, it still produced high error, likely due to setup inconsistencies. XFoil performed well for lift at certain angles (especially around 10° - 15°) but struggled with drag prediction. FoilSim showed the largest error and variability which was extremely noticeable at higher angles. These findings show that simulation tools must be used with a thorough understanding of their capabilities and limitations.

Conclusions

Overall, the accuracy of the 2D models at predicting the lift and drag was greater than expected, and both the single most accurate individual trial prediction across all data collected and the most accurate average error for a specified angle came from XFoil, with an individual error of 0.38% and average of 3.66% both at 15° . Generally, the ANSYS models underperformed at predicting lift, producing consistent

error between 70 and 90%. What's worth noting is that ANSYS' error was very consistent across all trials, suggesting it may be systemic, and although it cannot be ignored, it can be treated with a greater level of skepticism. Comparing the precision of the error, ANSYS generally seemed to produce results that followed more distinguishable, linear trends. We decided that, despite the accuracy of the 2D solvers, they do not show enough consistency, and the consistency that ANSYS showed coupled with that it likely had issues with its setups that were not discerned leads us to believe that, when tuned properly, it is likely the superior simulator when compared to the others. Further evaluations with greater oversight would be required to come to a definitive conclusion, however.

If conducted again, it may be preferable to use the Fluent solver that ANSYS offers. The solver used for this one is still an advanced fluid simulation, but rather than requiring computing of an inflated mesh, then transfer of that mesh into ANSYS' Fluent solver for computing of the fluid flow, it is capable of doing it natively within seconds. This affected the choice in using the proprietary integrated Discovery solver over ANSYS Fluent. The generation of each mesh and Fluent simulation for a 3D simulation requires roughly 30 minutes for each mesh and each simulation, and at 30 simulations, that would require a minimum of 30 hours of computational time alone to complete. Given the relative simplicity of our simulation, and the time constraints of the project, it was determined that the integrated solver would produce similar enough results and save a significant amount of time when compared to the more advanced Fluent solver, which is why it was ultimately used instead. This potentially could have contributed to whatever error was developed when compared to the real world data, as although it's expected to be an accurate simulation, Fluent is known to be the better of the two, and a more comprehensive analysis of ANSYS's software suite would use the Fluent solver.

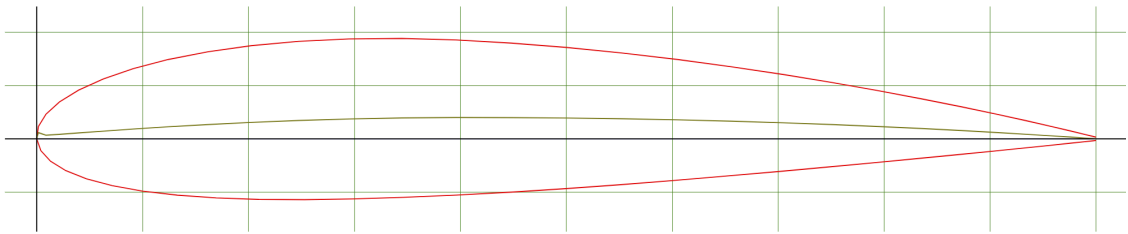
There is also a significant knowledge gap that cannot go unacknowledged when it comes to ANSYS. These results were produced with zero prior experience in any ANSYS software or computational fluid dynamic solvers in general, and inconsistencies in the setups or slight alterings of the enclosure between trials may have occurred. The setups were created using what knowledge could be coalesced during the duration of this report being developed, and may themselves have errors we did not

know enough to predict or prevent. The only oversight over this was very minute amounts of guidance from others who have used the more comprehensive solvers ANSYS offers.

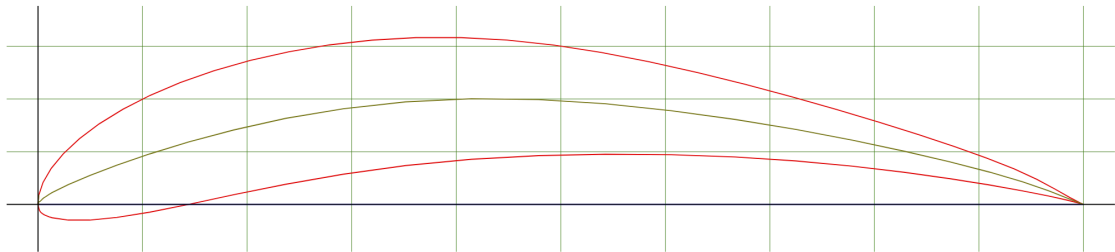
Appendix

A-1) NACA 2415 & E243 Splines

The NACA 2415 and E243 airfoil splines were sourced online via [AirfoilTools.com](https://www.airfoiltools.com) in the Lednicer and Selig data formats. This allowed them to be redrawn based on their spline points to create 2D and 3D models for our simulations and physical testing.



[NACA 2415 Spline.](#)



[Eppler 243 Spline.](#)

Works Cited

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